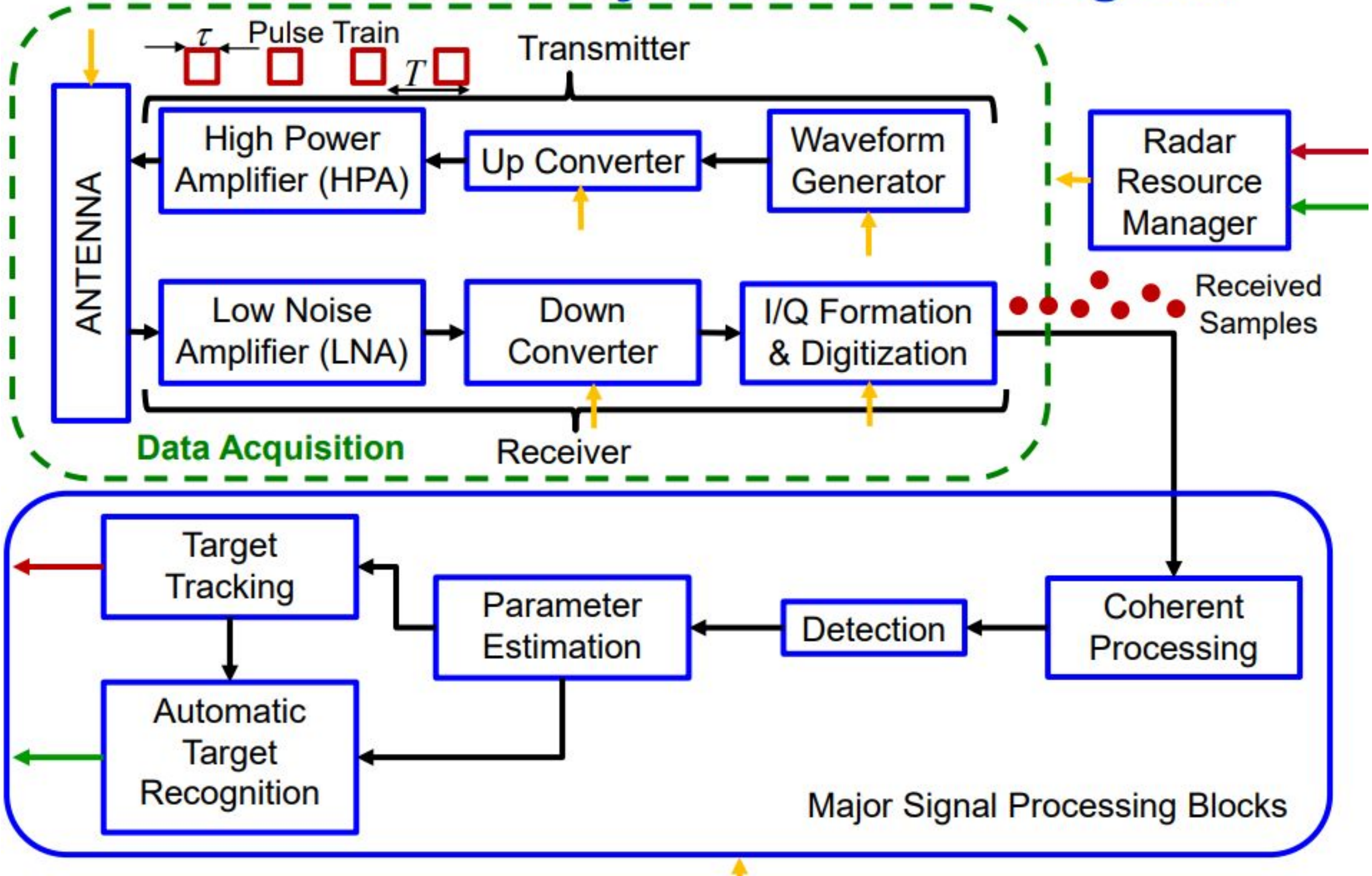
FRSP Lecture 2

Pulsed Radar Data Acquisition

EE417 / EE517
CUA Spring 2024

Tuesdays 7:00-9:30pm
Instructor: Prof. Kevin Russo

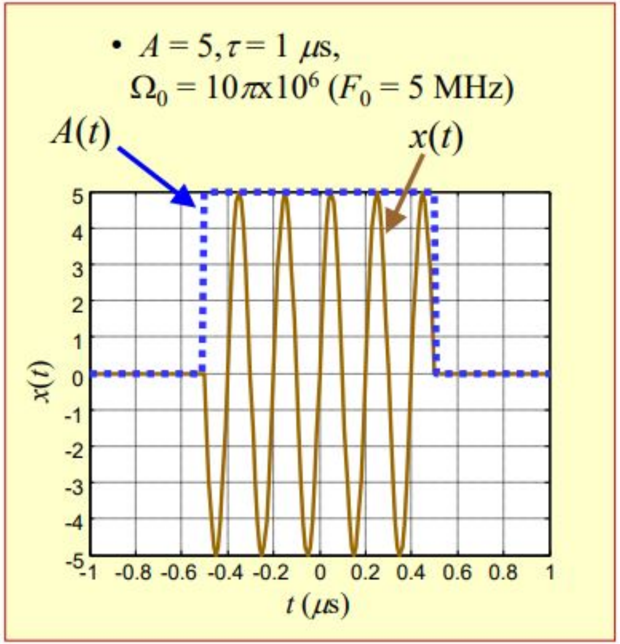
Generic Radar System Block Diagram



Narrowband Radar Pulse

- **Radar waveforms are usually**
  - **Bandpass,**
  - **Narrowband, and**
  - **Phase- or frequency-modulated**
- **The echo from a point scatterer has the same characteristics, so it is of the form**

$$\boxed{\bar{r}(t) = A(t)\sin\left[\Omega_0 t + \theta(t)\right]}$$

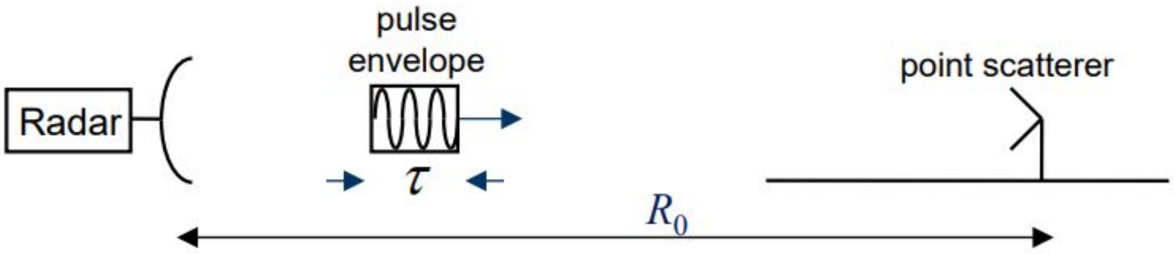


Ω=2πF

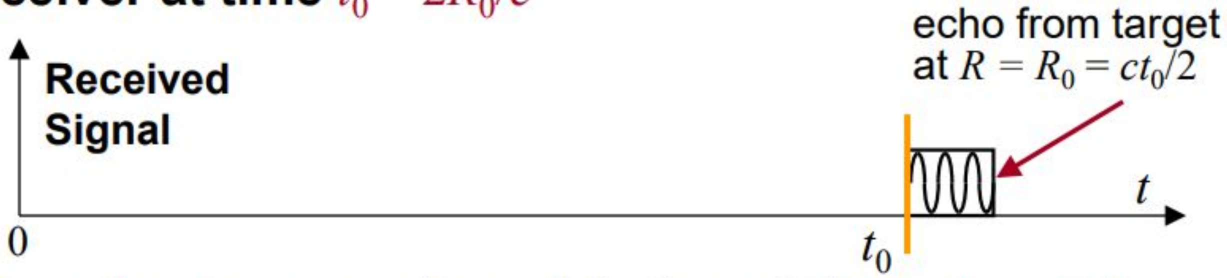
  - **(*overbar indicates a NON-baseband signal*)**
- **We want to measure the envelope $A(t)$ and the phase $\theta(t)$**
  - Envelope proportional to scatterer “size”
  - Phase will turn out to describe scatterer motion

Range is Equivalent to Time

- **Consider a pulse of length τ**
  - **Assume the leading edge is transmitted at time $t = 0$**
- **A point scatterer is located at range R_0:**



- **The leading edge of the echo will arrive at the receiver at time $t_0 = 2R_0/c$**



- **We estimate range from this time delay:** $R_0 = ct_0/2$

Range and Phase - 1

- **The transmitted RF pulse is** $\boxed{\bar{x}(t)} = A(t)\boxed{\sin(2\pi F t)}$
- **The echo from a scatterer R_0 meters away is**

Note: $\lambda = \dfrac{c}{F}$

$$
\begin{aligned}
\bar{y}(t) &= \boxed{\bar{x}(t - 2R_0/c)} \\
&= A(t - 2R_0/c)\sin\left[2\pi F(t - 2R_0/c)\right] \\
&= A(t - 2R_0/c)\sin\left[2\pi F t - 2\pi(2R_0 F/c)\right] \\
&= A(t - 2R_0/c)\boxed{\sin\left[2\pi F t - 2\pi(2R_0/\lambda)\right]}
\end{aligned}
$$

- **So compared to the transmitted pulse, the return echo is**
  - **Delayed by $2R_0/c$, and**
  - **Phase-shifted by $-2\pi(2R_0/\lambda) = -(4\pi/\lambda)R_0$ radians**

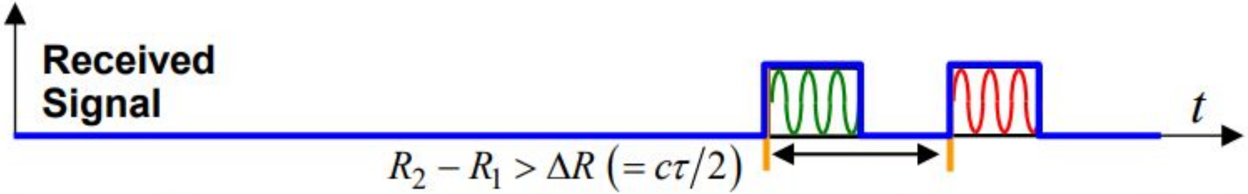
Range and Phase - 2

$$\bar{y}(t) = A(t - 2R_0/c)\boxed{\sin\left[2\pi Ft - (4\pi/\lambda)R_0\right]}$$

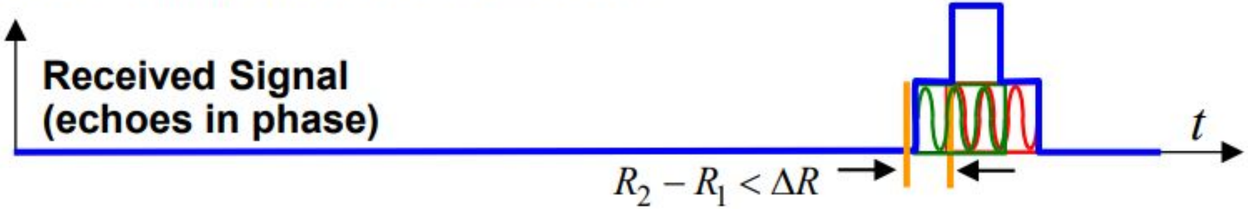
- **The return echo is**
  - **Delayed by** $2R_0/c$**, and**
  - **Phase-shifted by** $-(4\pi/\lambda)R_0$ **radians**
- **So, the echo phase shift is**
  - **Proportional to target range**
  - **Sensitive to sub-wavelength range changes**
- **Example:** $\sin\left[2\pi Ft - (4\pi/\lambda)(R_0 + \Delta R)\right]$
  - **Range changes by** $-\lambda/4$ m → $\Delta R = -\lambda/4$
    - The two-way path length changes by $-\lambda/2$ meters
  - **Phase of the echo changes by** $(-4\pi/\lambda)(-\lambda/4) = +\pi$ **rads (+180°)**
- **These phase shifts are used to measure Doppler, do SAR, do STAP, *etc.***

Resolving Two Scatterers in Range

- Two scatterers separated by more than $\Delta R = c\tau/2$ meters provide two distinct, nonoverlapping echoes at the receiver output:



- Two scatterers separated by less than ΔR in range overlap to form a single, unresolved echo:



- The minimum separation to avoid overlap, $\Delta R = c\tau/2$, is called the range resolution
- Later we will see that modulated waveforms follow a different rule

Example: Range Resolution for a Simple Pulse

- If the pulse length is 10 μs, what is the range resolution?
  - $\Delta R = c\tau/2 = (3\times10^{8})(10\times10^{-6})/2 = 1500 \text{ m} = 1.5 \text{ km}$

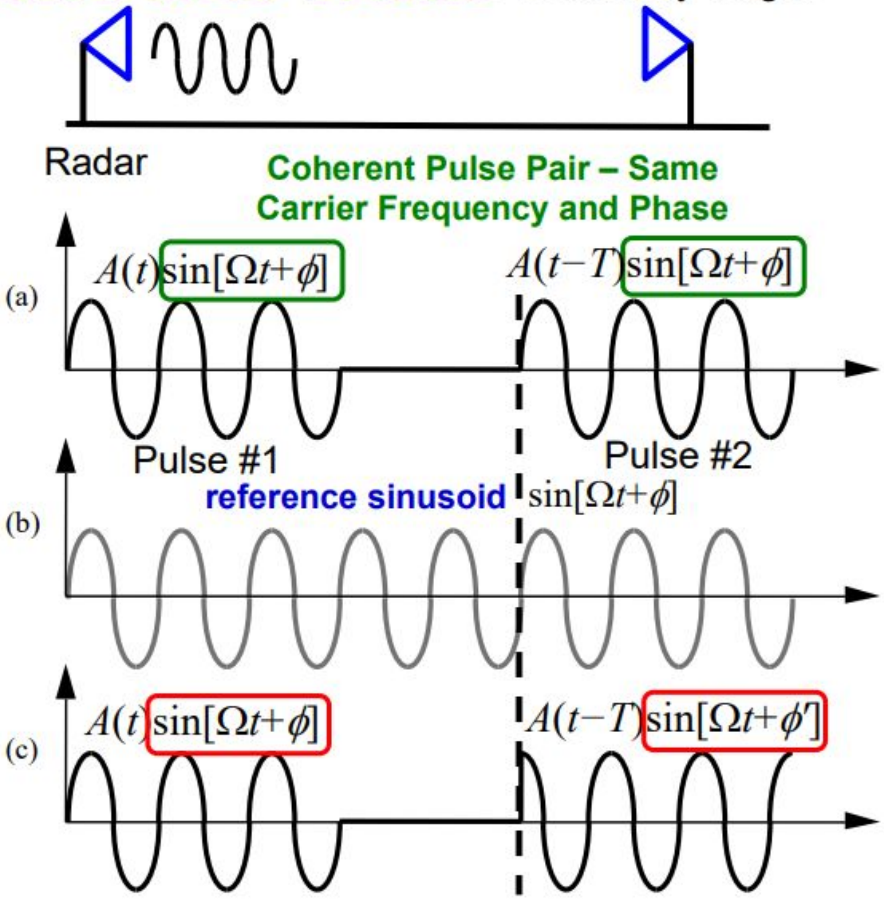
Pulse Width	Range Resolution (m)
1 nanosecond (1E-9)	0.15 (0.00015 km)
1 microsecond (1E-6)	150 (0.15 km)
1 millisecond (1E-3)	150,000 (150 km)

- If we want to achieve range resolution of 3 m, what would the pulse length have to be?
  - $\tau = 2\Delta R/c = 2(3)/(3\times10^{8}) = 2\times10^{-8} \text{ s} = 20 \text{ ns}$

Measuring Phase and the Complex Sinusoidal Signal Model

Coherent Pulse Trains

- "Coherent processing" over multiple pulses is required in many algorithms
  - So that pulse-to-pulse phase changes are due to radar-target motion, not to changing or unknown transmitted phases
  - Ultimately, so data from different pulses can be successfully combined
- Implies that the signal phases are referred to a common, stable oscillator
- Implies a need to measure the phase of pulse echoes



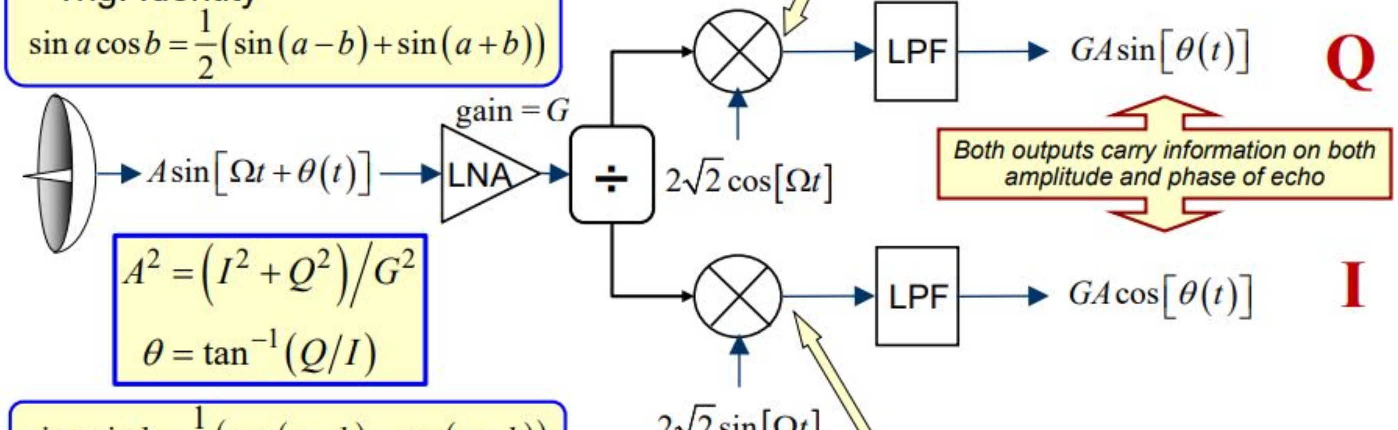
Noncoherent Pulse Pair - Same Carrier Frequency, Different Phase

Quadrature or "I/Q" Receiver

$$2\sqrt{2}G\cos[\Omega t]\cdot\frac{A(t)}{\sqrt{2}}\sin[\Omega t+\theta(t)]=GA(t)\sin[\theta(t)]+GA(t)\sin[2\Omega t+\theta(t)]$$

Trig. Identity

$$\sin a\cos b=\frac{1}{2}(\sin(a-b)+\sin(a+b))$$



$$A^2=\left(I^2+Q^2\right)/G^2$$

$$\theta=\tan^{-1}(Q/I)$$

$$\sin a\sin b=\frac{1}{2}(\cos(a-b)-\cos(a+b))$$

$$2\sqrt{2}G\sin[\Omega t]\cdot\frac{A(t)}{\sqrt{2}}\sin[\Omega t+\theta(t)]=GA(t)\cos[\theta(t)]-GA(t)\cos[2\Omega t+\theta(t)]$$

- **Each channel, by itself, is called a synchronous detector**
- **The lower channel (proportional to $\cos\theta$) is the in-phase ("I") and the upper (proportional to $\sin\theta$) is the quadrature ("Q")**
- **Allows measurement of both amplitude A and phase θ of echo**

Complex Signal & Equivalent Receiver

- **We can create a complex-valued baseband signal using the I and Q signals as the real and imaginary parts:**

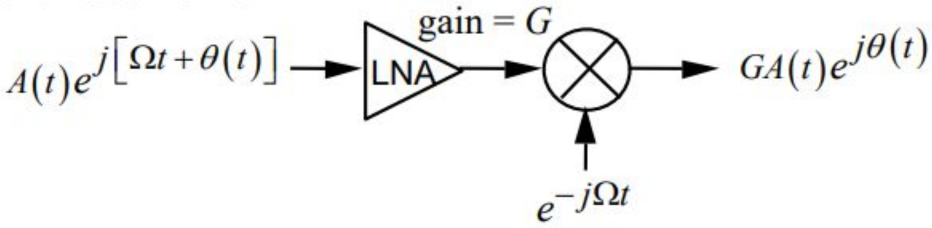
$$y(t) \equiv I(t) + jQ(t)$$

$$= GA(t)\{\cos[\theta(t)] + j\sin[\theta(t)]\} = GA(t)e^{j\theta(t)}$$

  - **Euler's formula:**

$$\exp(j\theta) = \cos\theta + j\sin\theta$$

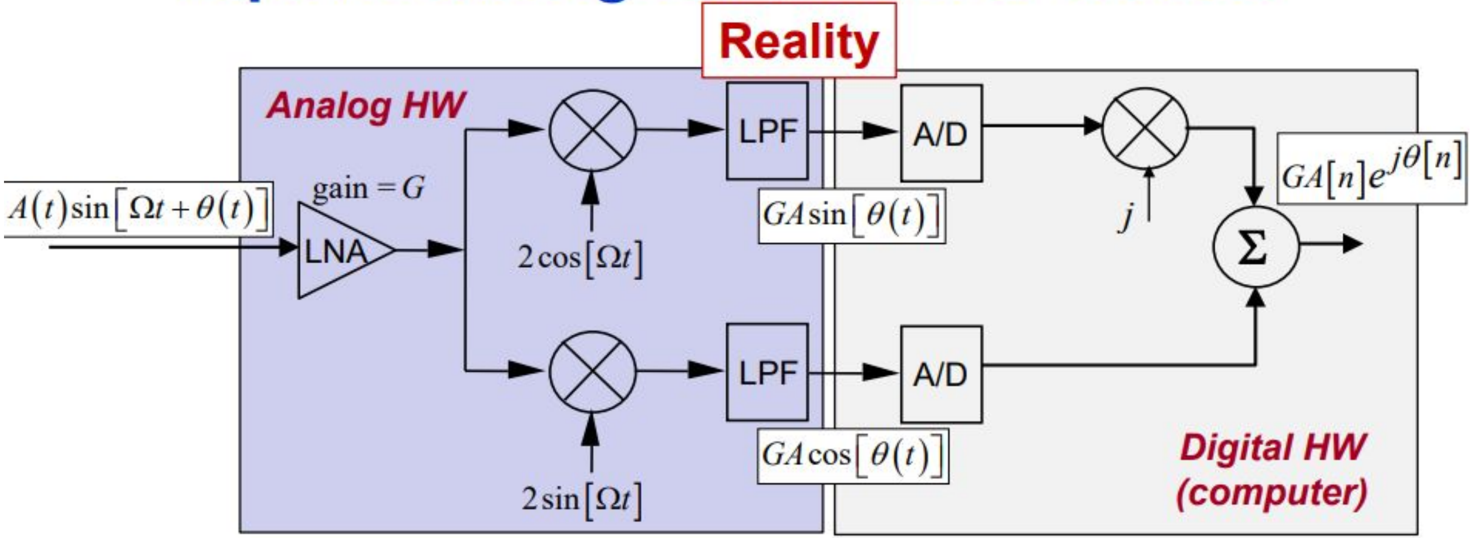
- **We actually record and act on complex data in our signal processor**
- **An equivalent mathematical model of the received signal and the quadrature receiver is**

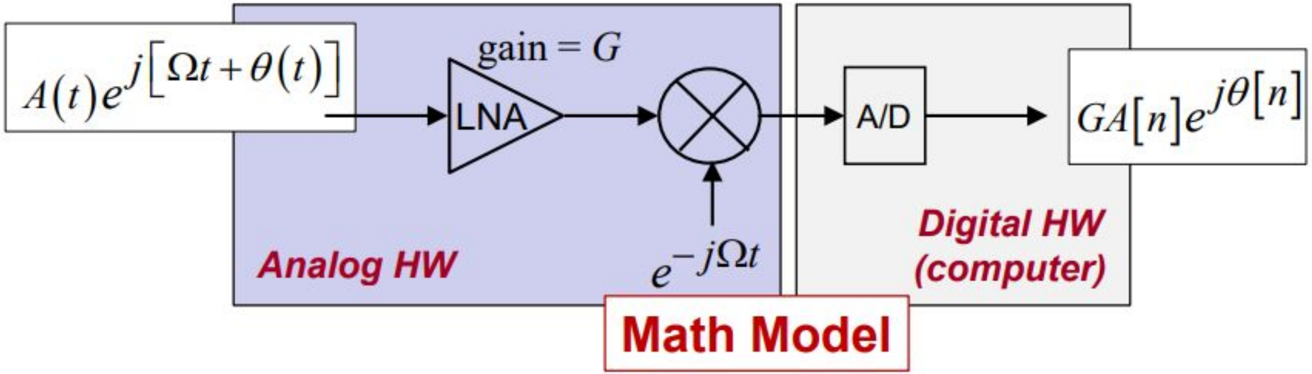


Same Result!

- **Complex exponential signal models are much easier to manipulate than trigonometric function models, so that's what we'll use**

Equivalent Signal+Receiver Models

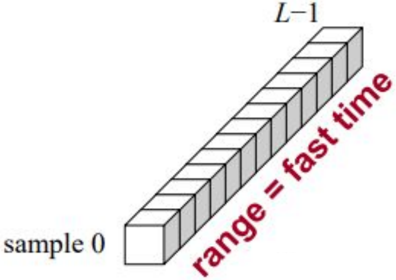




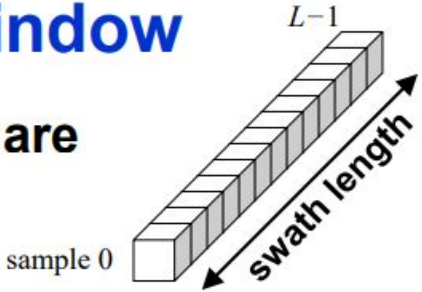
Pulsed Radar Data: Organization, Sampling Rates, Ambiguities, and Blind Zones

Collecting Pulsed Radar Data: 1 Pulse, Multiple Range Samples

- When using a coherent receiver, each range sample comprises one "I" sample and one "Q" sample, forming one **complex** number $I+jQ$
- So each **range bin** is 1 complex number stored in your signal processor memory
  - Also called **range gates**, **range cells**, or **fast time samples**
- Each range bin represents the complex echo amplitude from a different range interval of extent $c\tau/2$ (more generally, ΔR) meters
- Samples are usually taken at about 1 resolution cell spacing or a little less, but not always



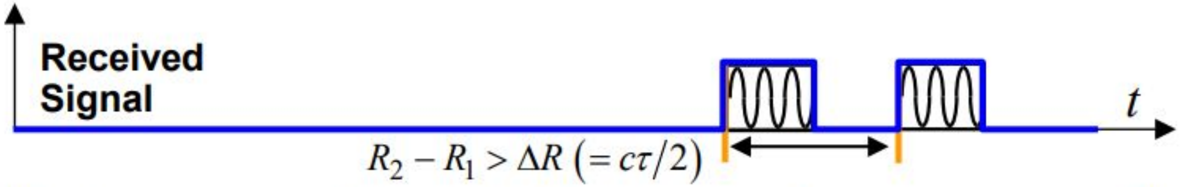
Range Swath or Range Window



- The swath is the interval of ranges that are sampled for a single transmitted pulse
  - Also called swath length or depth
  - Range window may also be used instead of range swath
- What do I do if I want my range swath to cover ranges of 10 km to 100 km?
  - Don't start taking samples until you've waited long enough to get the echo from 10 km away …
    - So wait $2(10{,}000)/3\times10^8 = 67\ \mu s$ to start taking samples after the pulse is transmitted
  - … and stop when you have taken samples long enough to get the echo from 100 km away
    - So stop taking samples $670\ \mu s$ after the pulse is transmitted

How Fast Should We Sample in Range?

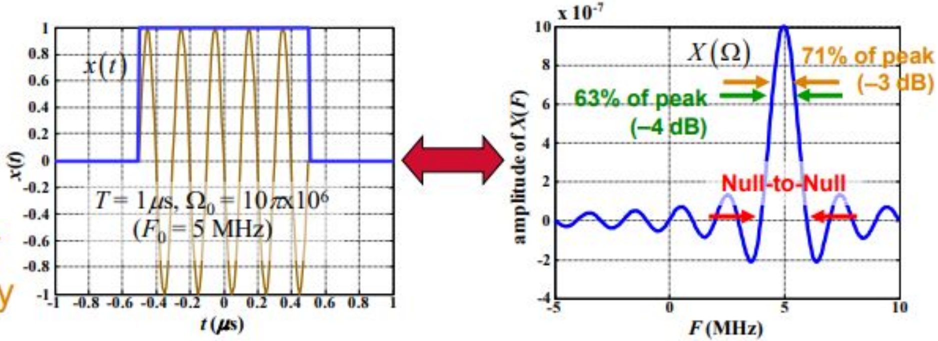
- One answer is to not leave any “gaps”:
  - We don’t want to miss sampling a target return



- That suggests we take samples no more than τ seconds ($c\,\tau/2$ m) apart
- Another answer is to apply the Nyquist theorem, but then we need to know the bandwidth B of the fast-time data
  - Nyquist theorem defines minimum sampling rate as the waveform bandwidth
- So what is the pulse bandwidth?

Nyquist Sampling Rate in Fast Time

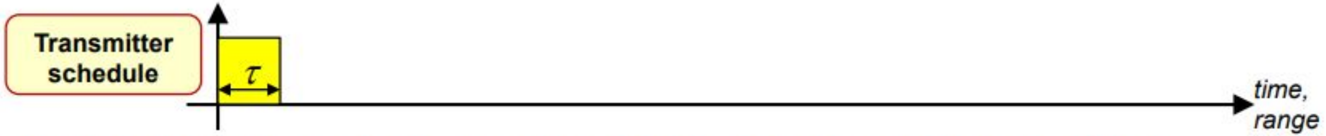
- **For a constant-frequency square pulse, the …**
  - **2-sided 3 dB bandwidth is $0.89/\tau$**
    - 72% of total energy in sinc[2] pattern



  - **2-sided 4 dB bandwidth is $1/\tau$**
    - 78% of total energy in sinc[2] pattern
    - (Also happens to equal the Rayleigh (peak-to-null) and RMS bandwidths)
  - **Null-to-null bandwidth is $2/\tau$**
    - 91% of total energy in sinc[2] pattern
- **The (two-sided) "bandwidth" is often taken as about $1/\tau$ Hz**
  - **This becomes the Nyquist sampling rate**
- **So we need to sample at a rate of $1/\tau$ samples/sec**
  - **Same as one sample every τ seconds ($c\tau/2$ m)**
  - **Same conclusion we got from spacing of pulse echoes**
- **It is common to increase this rate by 20% or so**

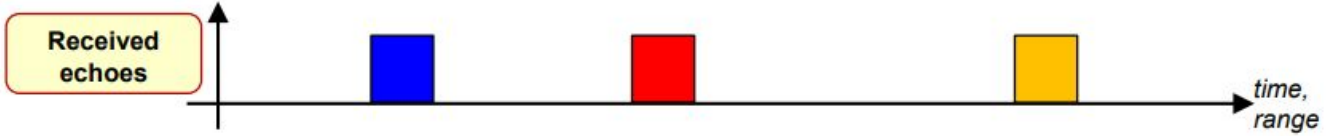
Range Blind Zones: Eclipsing: 1 Pulse

- Consider a single transmitted pulse:



- A monostatic radar cannot receive while a pulse is being transmitted due to transmitter leakage, so the receiver is isolated from the antenna and transmitter during pulse transmission:

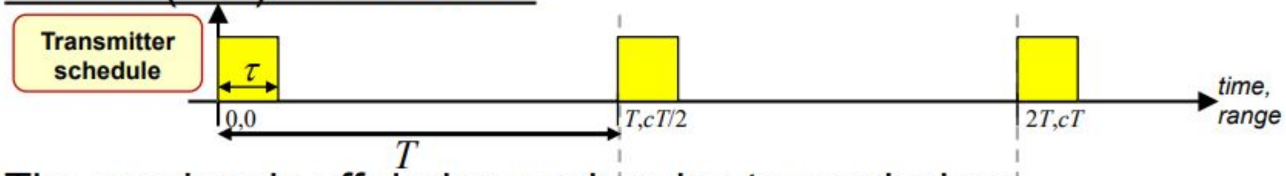
- Suppose there are three targets at various ranges. All three target echoes are observed in the receiver:



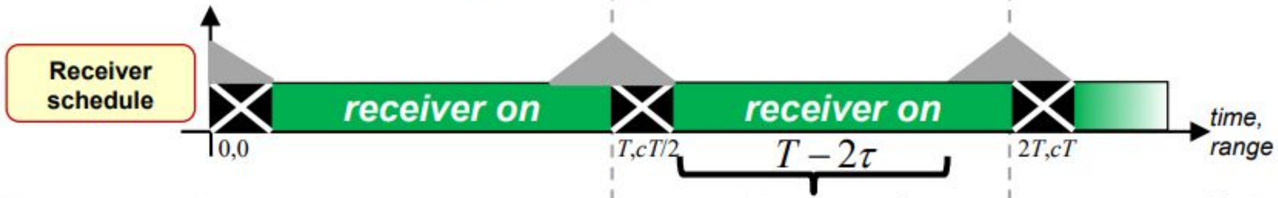
- **So far, so good. What if we send some more pulses?**

Range Blind Zones: Eclipsing: 3 Pulses

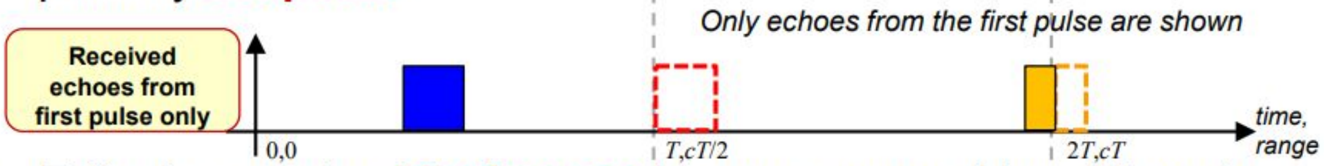
- Now suppose additional pulses are transmitted at a <u>pulse repetition interval (PRI) of T seconds</u>:



- The receiver is off during each pulse transmission:

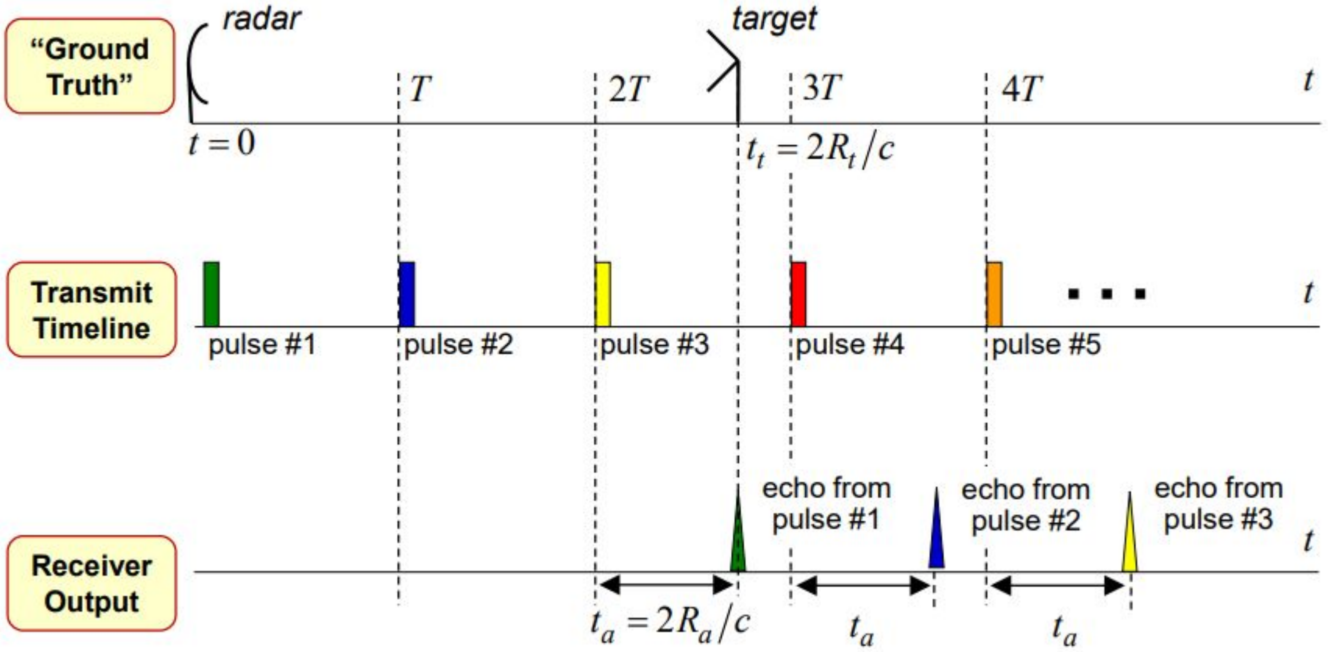


- Target echoes at ranges equal to multiples of $cT/2 \pm c\tau/2$ are fully or partially **eclipsed**:



- Eclipsing creates **blind zones** in range, spaced by $cT/2$ meters
  - The receiver "off" period may be lengthened to allow for switching time and decay of strong near-in clutter

Range Ambiguity & Foldover



- Once steady state is reached (pulse #3 and later), the target appears to be at a shorter *apparent* range R_a after every pulse
  - I.e., it "**folds over**" into the range interval $(0,cT/2)$
- Many systems only measure R_a, so the target range is now **ambiguous**
  - Is it really R_a, or maybe $R_a + cT/2$, or $R_a + 2cT/2$, or …?

Unambiguous Range

- **Unambiguous range R_{ua} is the …**
  - Maximum range a pulse can get to and still get back before the next pulse is transmitted

$$R_{ua} = \frac{c \cdot T}{2} = \frac{c \cdot PRI}{2} = \frac{c}{2PRF}$$

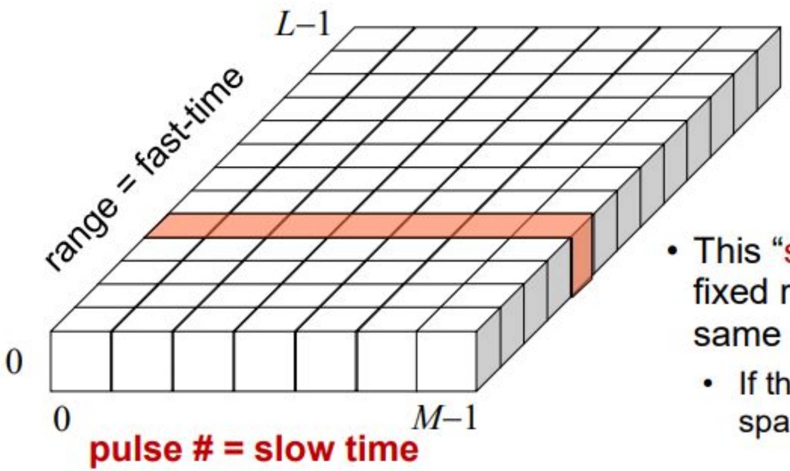
  - Maximum range of a target before it starts to fold over
- **Range foldover also occurs with clutter**
- **The maximum number of range ambiguities or range foldovers depends on R_{ua} and the maximum range of detectable target and clutter echoes**
  - Successive intervals of length $cT/2 = R_{ua}$ overlay one another in steady state

$$R_a = R_0 + m\frac{cT}{2} \quad m = \pm 1, 2, 3, \ldots$$

- **Low PRF → large R_{ua} → Range ambiguity is less of an issue**

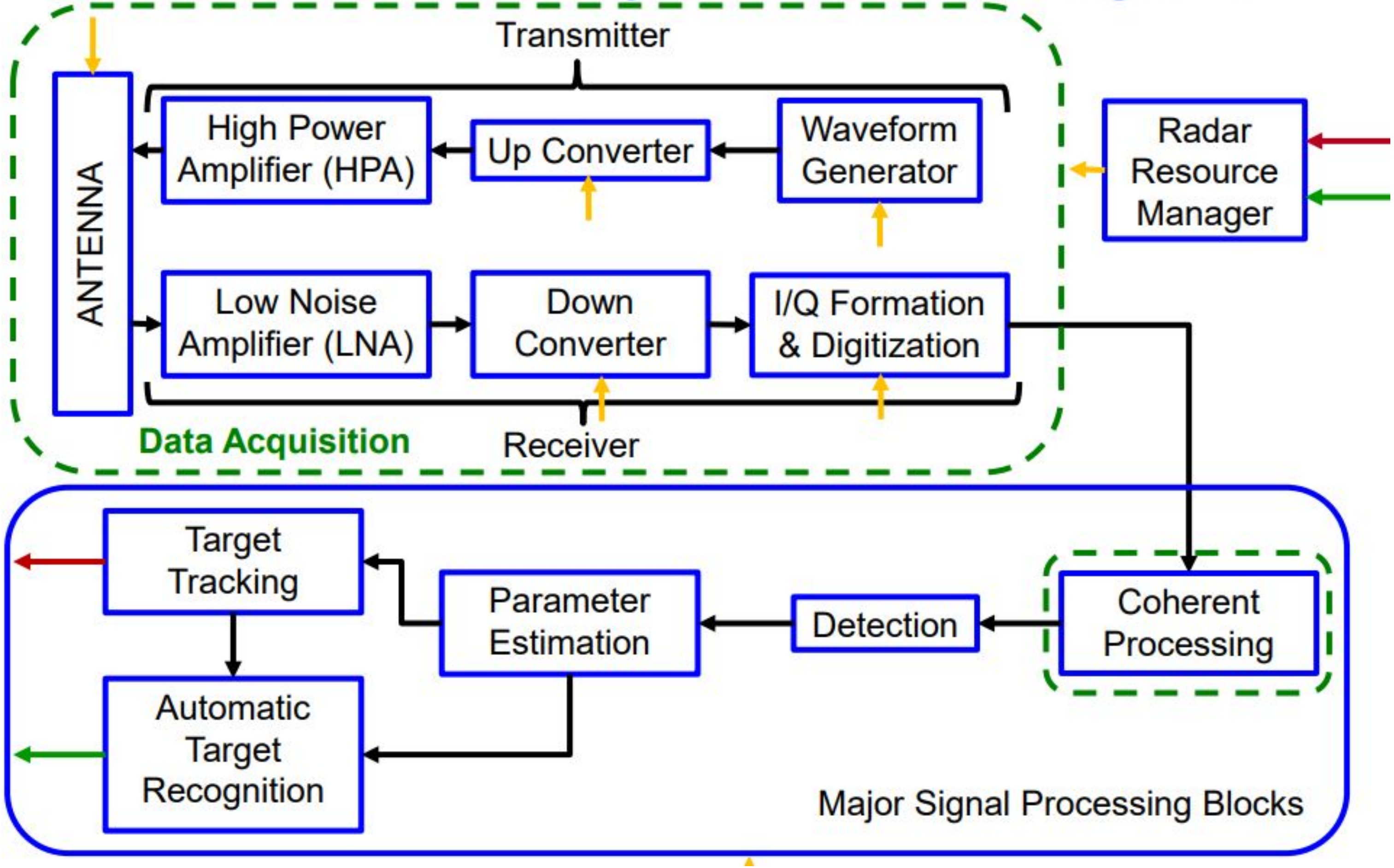
Collecting Pulsed Radar Data: Multiple Pulses

- Repeat for multiple pulses in a "coherent processing interval" (CPI)
- The new axis is pulse number, also called slow time
- The time between pulses is the pulse repetition interval (PRI) or interpulse period (IPP)
- "CPI" *usually* implies the same radar frequency, same pulse waveform, and constant PRF for all pulses in the CPI (but not always)
- This matrix ("CPI") of data is often the basic unit for processing



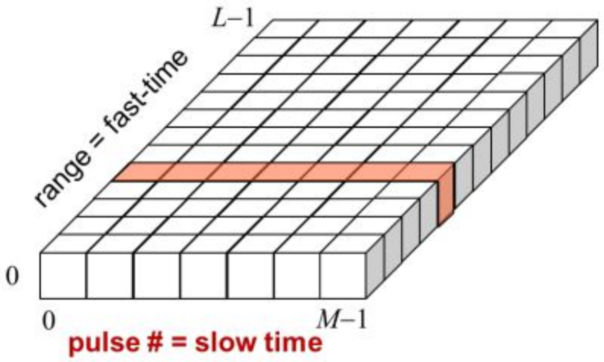
- This "slow-time" sequence of samples for a fixed range bin represents echoes from the same range interval over a period of time
  - If the antenna illuminates the same region in space for the duration of the CPI

Generic Radar System Block Diagram



Sampling in Slow Time: How Do We Choose the PRF?

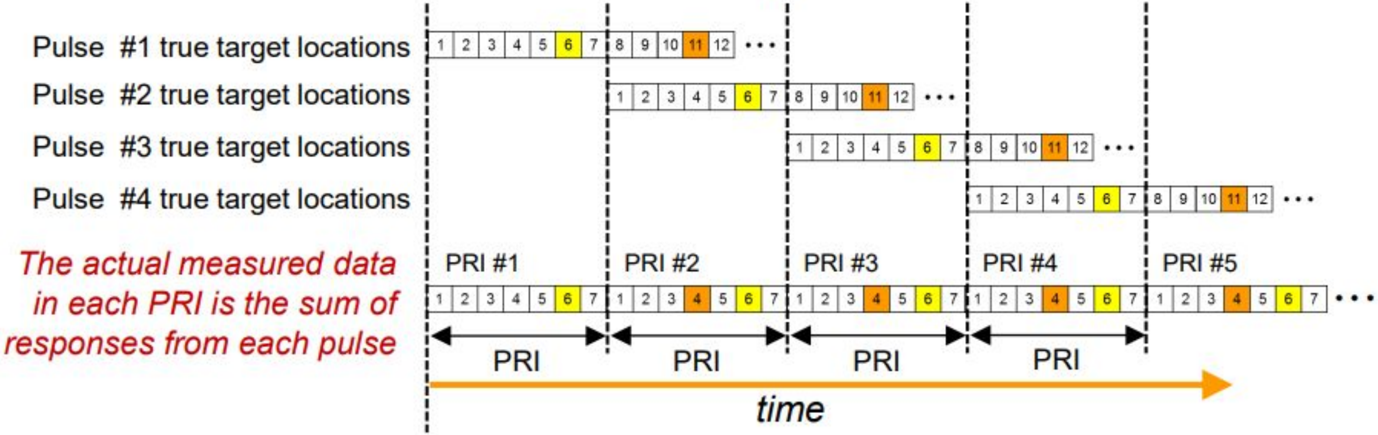
- **The time between pulses is the pulse repetition interval (PRI) or interpulse period (IPP)**
- **Therefore the slow time sampling rate is the pulse repetition frequency (PRF) of the radar**
  - $PRF = 1/PRI$
- **To satisfy the Nyquist requirement, the PRF must be greater than the expected bandwidth of the slow-time data sequence**
  - Not always possible or even desirable
- **That bandwidth is the spread in Doppler frequency of the scatterers in the range bin of interest**
- **The Doppler bandwidth depends on**
  - Radar frequency
  - Radar-target or radar-clutter geometry
    - Opening/closing, cone angle
  - Radar platform speed
  - Target speed (hard targets)
  - Clutter spectrum width (clutter)
  - Antenna beamwidth (clutter)
  - ***More on this later …***



Fast Time/Slow Time Matrix with Two Targets & Range Foldover - 1

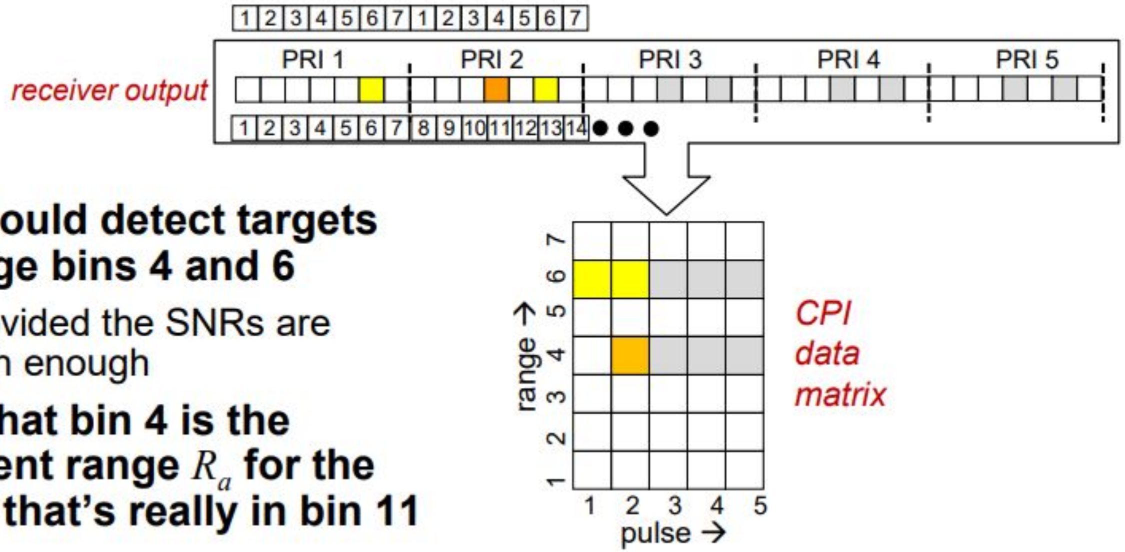
- **Suppose we have two targets at true ranges R_t corresponding to range bins 6 and 11**

- **Measure with a PRI corresponding to 7 range bins**
- **Then the received data will appear like this:**



Fast Time/Slow Time Matrix with Two Targets & Range Foldover - 2

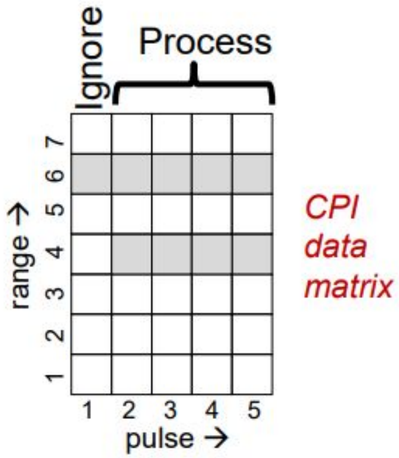
- **Pulse-Doppler processing assembles the PRI data into a range/pulse number (fast time/slow time) matrix for this CPI**



- **We should detect targets in range bins 4 and 6**
  - Provided the SNRs are high enough
- **Note that bin 4 is the apparent range R_a for the target that's really in bin 11**

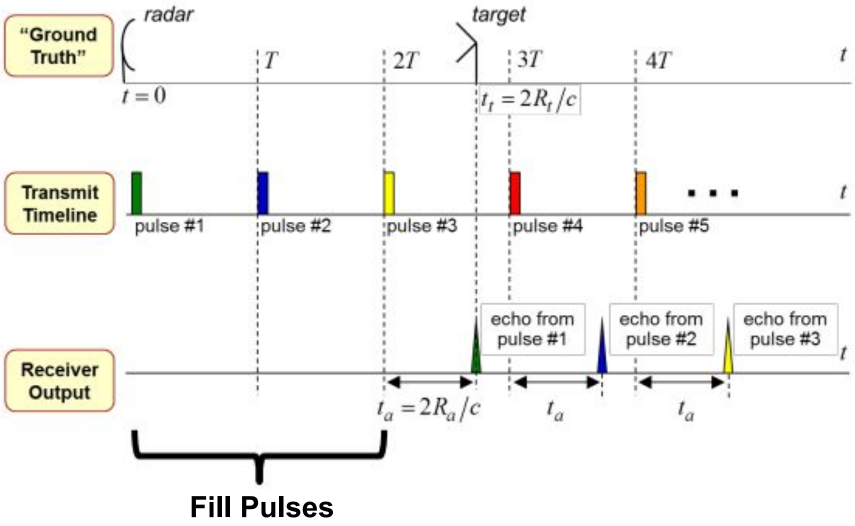
- ***We'll see later how to determine the real ranges …***

Fill Pulses



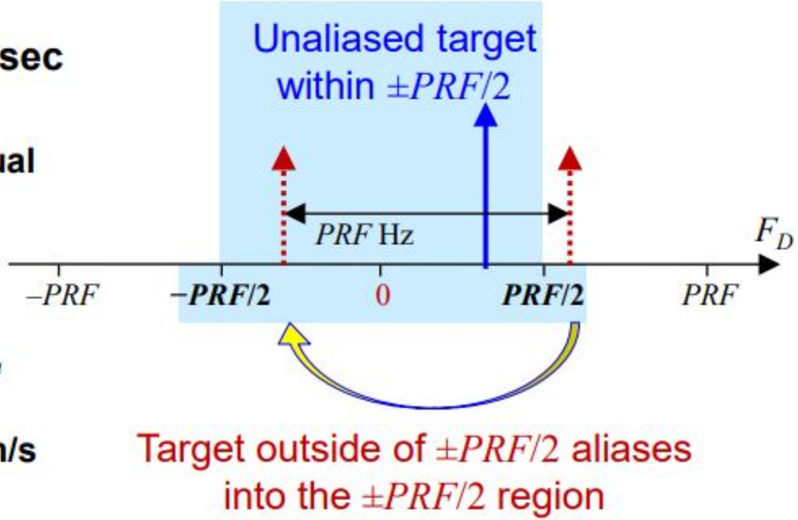
- **If our maximum-range target is between 1 and 2 PRIs out …**
  - **We won't achieve a steady-state data pattern until pulse #2, so …**
  - **Pulse #1 is a "fill" pulse**

- **Earlier range ambiguity example had 2 fill pulses**
- **Fill pulses ensure a full set of pulses on a range ambiguous target as well as long range clutter**



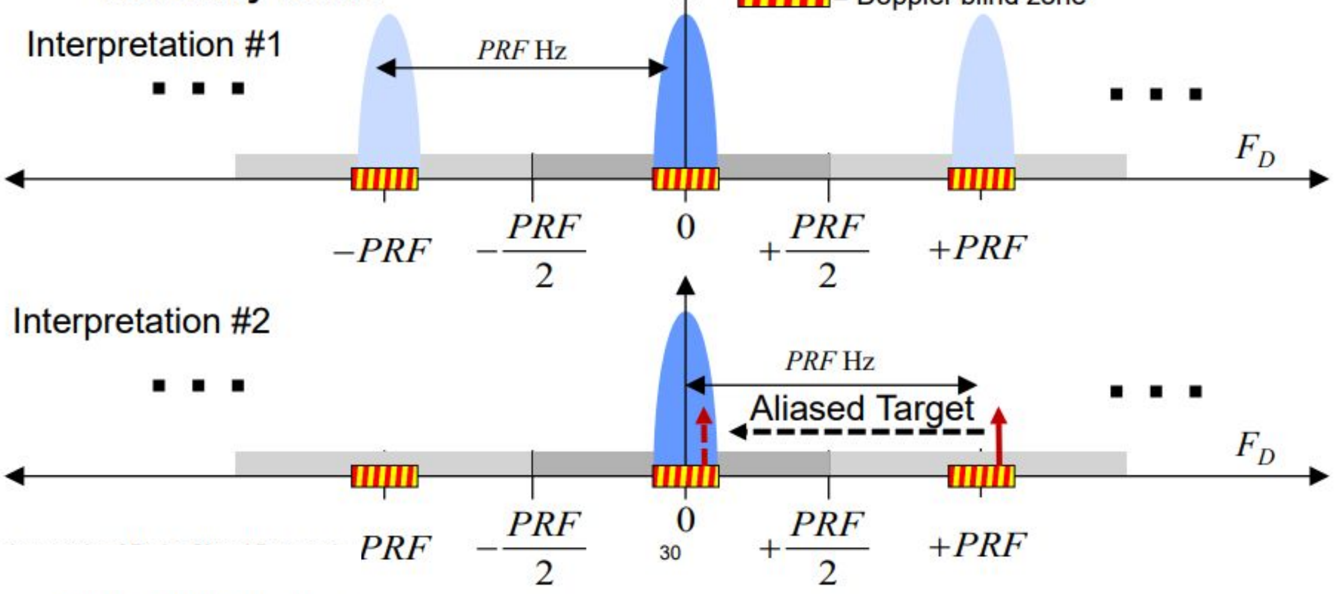
Doppler Spectrum Aliasing

- **The Doppler spectrum is usually obtained by computing the 1D Fourier transform in the slow-time dimension**
- **The slow-time sampling rate F_s is the PRF of the radar, $F_s = PRF$**
- **A fact from DSP: frequencies outside the range $\pm PRF/2$ samples/sec alias into that range**
  - **Apparent Doppler differs from actual**
  - **Only the "principal period" from $-PRF/2$ to $+PRF/2$ Hz is processed**
  - **The width of that interval is the unambiguous Doppler interval F_{Dua}**
  - **The corresponding velocity ambiguity interval is $v_{ua} = \lambda \cdot F_{Dua}/2$ m/s**
- **High PRF → large v_{ua} or F_{Dua} → velocity or Doppler ambiguity is less of an issue**
- ***We'll see later how to determine the real Doppler shifts or velocities …***



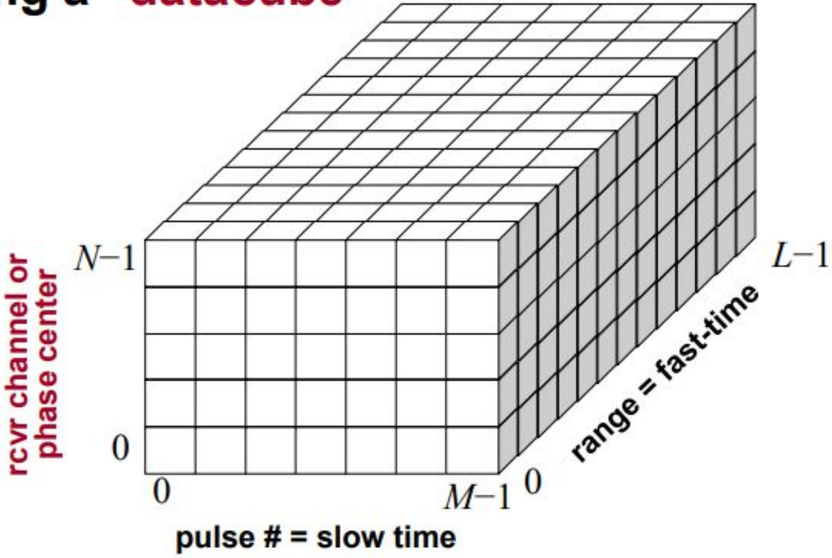
Doppler Blind Zones

- Strong clutter and aliasing create repeating blind zones in Doppler
  - Targets are likely undetectable due to poor signal-to-clutter ratio
  - Interpretation #1: Periodic Doppler spectrum repetition of the clutter creates blind zones in Doppler at intervals of PRF Hz
  - Interpretation #2: Alternatively, targets around multiples of the PRF alias into the clutter around zero Doppler
- Notional effective spectrum of noise + mainlobe ground clutter for a stationary radar:



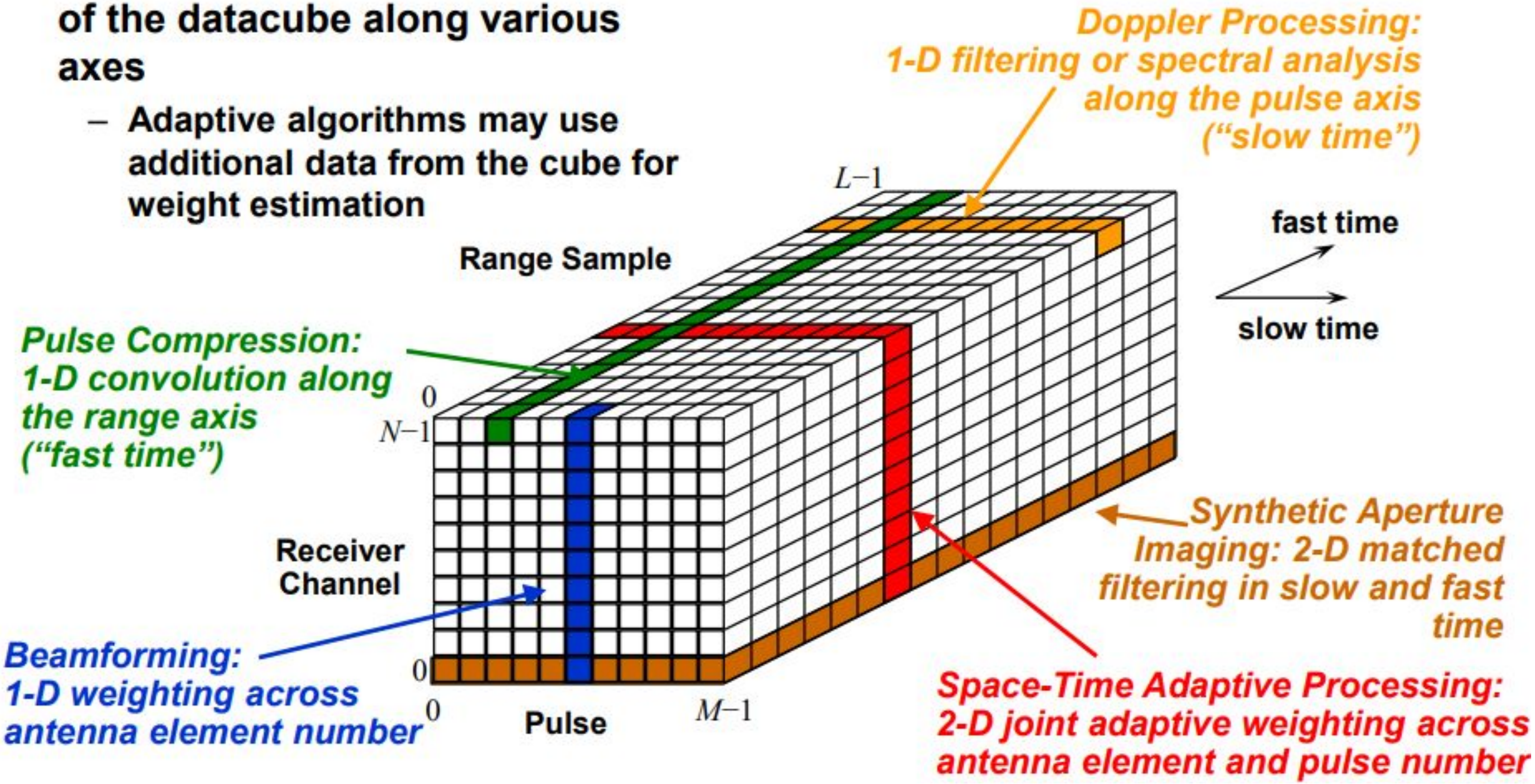
Collecting Pulsed Radar Data: Multiple Receiver Channels

- **If** we have multiple receiver channels …
  - Multiple phase center antenna, monopulse antenna, *etc.*
- … the entire fast/slow time matrix is collected for each receiver, …
- … forming a "**datacube**"



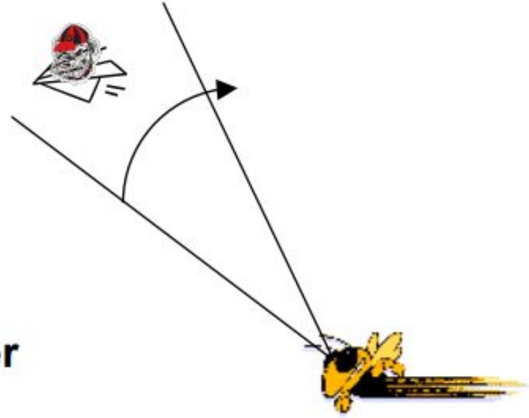
Processing the Radar Datacube

- Common radar signal processing algorithms act on 1- or 2-D slices of the datacube along various axes
  - Adaptive algorithms may use additional data from the cube for weight estimation



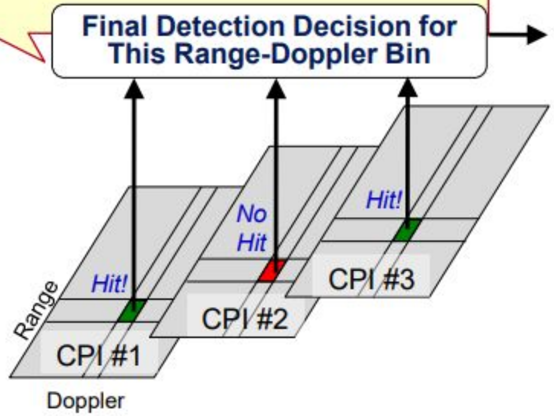
Multiple CPIs: Dwells

- Many radars will collect more than one CPI of data from the same region of space
- Example:
  - Radar with 3° beamwidth, scanning in azimuth at 60°/second
  - A target is within the main beam for 3/60 sec = 50 ms
  - Assume a 2 kHz PRF and 20 pulses per CPI → 10 ms per CPI
  - So we could collect up to 5 CPIs while the target is in the beam
    - Probably fewer, say 3 or 4 CPIs, with some dead time between
- We'll call the time a target is within the beam (staring or scanning) the dwell time
- WARNING: This terminology is not well-standardized:
  - "Dwell" might mean the same as "CPI" to some people; or
  - "Scans", "frames" might also refer to multiple CPIs; or maybe to multiple "dwells"!



Use of Multiple CPIs

- Each CPI is processed separately to improve its signal-to-interference ratio, maybe to do single-CPI detection
  - Jammer cancellation, clutter suppression, resolution enhancement, target detection
  - Raw CPI often converted to range-Doppler format first
- Data from multiple coherently-processed CPIs is then combined to provide higher-confidence decisions with fewer limitations, e.g.
  - Noncoherent integration across CPIs for improved detection performance (pre-threshold test)
  - "Binary integration" ("*M*-of-*N*" processing) for improved detection performance (post threshold test)
  - Range and Doppler ambiguity resolution
  - Blind zone mitigation
- Usually cannot combine data coherently across CPIs
  - Separate pulse trains for separate CPIs are not coherent with one another



Radar Data Time Scales

- **Radars collect and combine data over time scales covering several orders of magnitude**